

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifi-
8 cation demonstrating how cognitive architectures can emerge as at-
9 tractor states from established physical primitives, without introduc-
10 ing new fundamental physics. Starting from the Poincaré group of
11 spacetime symmetries and finite phase space, we construct a frame-
12 work for predictive processing. Spontaneous symmetry breaking gen-
13 erates necessary topological boundaries, and finite phase space sub-
14 jects these boundaries to Landauer’s thermodynamic erasure. Agency
15 arises when systems evolve along paths of least action, minimizing en-
16 tropy production via internal structural resonance with external per-

17 turbations. Scaling this dynamic to macroscopic levels aligns with
 18 emerging paradigms in neuroscience—specifically classical electrody-
 19 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
 20 tion, culminating in a topological fixed point: a unified “Self.” We
 21 state the claim narrowly, because the subject matter invites a wider
 22 reading than the work supports: no new physics is derived here. Every
 23 physical result the chain invokes is established—Picard–Lindelöf, Kib-
 24 ble’s classification of topological defects, Landauer’s erasure bound,
 25 Gibbs’ inequality, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s
 26 fixed point. What is new is that they are stated in one formal lan-
 27 guage, their consistency mechanically checked, and the points where
 28 the chain relies on stipulation rather than derivation located exactly.
 29 The development declares no axioms and every structure carrying
 30 physical content is inhabited by a witness; the synchronization step is
 31 proved from a dynamics for initial data inside an explicit basin and
 32 assumed outside it, and the identification of a global section of the
 33 probability sheaf with the unity of experience is a stipulation, marked
 34 as such throughout rather than absorbed into the derivation. The
 35 framework is nonetheless falsifiable in one place that is its own and
 36 no rival’s: recovery from sudden awakening should follow a delayed
 37 sigmoid rather than an exponential, because the variable gating it is
 38 astrocytic volume on a timescale of seconds to minutes rather than cor-
 39 tical phase realignment on a timescale of milliseconds. As a corollary,
 40 we show why rigid von Neumann architectures (e.g., GPUs) are ther-

41 modynamically disadvantaged or structurally constrained from reach-
42 ing these specific continuous thermodynamic attractors.

43 **1 Introduction**

44 The pursuit of a physical theory of consciousness typically involves search-
45 ing for the biological correlates of mind or treating subjective experience as
46 an abstract computational algorithm. This paper asks a more fundamental
47 question: If we assume only basic physical primitives—continuous symme-
48 tries (the Poincaré group), geometry, and thermodynamics—is the emergence
49 of a unified, perceiving entity a natural physical attractor?

50 We propose a framework where the structural properties of consciousness
51 are derived as physical attractor states. Establishing a deductive chain from
52 basic principles, we demonstrate how continuous symmetries break to form
53 boundaries, how those boundaries dissipate free energy as non-equilibrium
54 steady states, and how their evolution along paths of least action results
55 in physical resonance with their environment. This eventually scales into
56 the macroscopic electromagnetic unity proposed by emerging field theories
57 of mind. To avoid critiques of “math theatre,” we delineate between literal
58 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
59 boundaries from spontaneous symmetry breaking) and theoretical mappings,
60 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
61 spective aligned with Russellian Monism: objective measurements of macro-

62 scopic electromagnetic fields constitute the third-person “outside,” while the
63 system’s internal physical deformation to reach structural resonance consti-
64 tutes the first-person intrinsic experience.

65 **What is new here, and what is not**

66 A framework paper about consciousness invites a particular suspicion, and it
67 is better answered at the start than at the end. **No new physics is derived**
68 **here.** Every physical result the chain invokes is established, and most of
69 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
70 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
71 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
72 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative
73 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
74 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
75 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
76 Bell and Crooks relating dissipation to nonpredictive information (Still et
77 al., 2012); and Banach’s fixed-point theorem (Banach, 1922). A physicist
78 reading this paper will recognize all of it and would establish the point in
79 three paragraphs if we did not. Conceding it costs nothing.

80 What is new is of a different kind, and it is a smaller claim than a theory
81 of consciousness usually makes. These results have been stated in one formal
82 language, their mutual consistency has been mechanically checked, and the
83 places where the chain relies on stipulation rather than derivation have been

84 located exactly. The value of a machine-checked chain is not that any link is
85 novel; it is that no link can quietly borrow from another, that a hypothesis
86 cannot be smuggled in as a definition, and that the seams — where physics
87 stops and interpretation begins — are fixed in the source rather than nego-
88 tiable in the prose. Section 2.1 reports what the check cost: three of five
89 original axioms each proved **False**, from one shared root cause, and none of
90 the three had been caught by reading.

91 The framework is not thereby unfalsifiable, and one of its predictions is
92 its own rather than a rival's. Section 11 argues that recovery from sudden
93 awakening should follow a *delayed sigmoid* rather than an exponential, be-
94 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
95 clearance, on a timescale of seconds to minutes, while the cortical rhythms
96 whose realignment is usually taken to constitute the recovery move in mil-
97 liseconds. That is a claim about functional form; it is fittable against data of
98 a kind already collected; and a purely electrical account of the same recovery
99 predicts a different form.

100 **2 Formalization in Lean 4**

101 A core contribution of this work is the verification of our deductive chain
102 using the Lean 4 interactive theorem prover. This approach builds on recent
103 efforts to formalize consciousness and brain semantics, such as the 'OnlyOne'
104 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,

2026). It is necessary to explicitly distinguish which claims have been math-
 ematically derived as theorems within the proof assistant, and which remain
 irreducible physical postulates. Table 1 maps the primary claims of this
 paper to their current formalization status.

Table 1: Mapping of physical claims to Lean 4 formalization status. The development declares **no axioms**: `#print axioms` on any result above reports only `propext`, `Classical.choice` and `Quot.sound`. Earlier drafts carried five axioms, three of them inconsistent; all five became class fields, so a postulate is an obligation on each instance rather than a global assertion (Section 2.1), and two of the five have since been eliminated entirely (Derivation 5). “Theorem” still does not mean “physically established”: the postulates live in hypotheses and instance obligations, and every such class is inhabited by a witness in `Examples.lean`, so no row is vacuous.

Conceptual Claim	Lean 4 Implementation	Status
Pointwise Vacuum Minimization	<code>pointwise_vacuum_of_global_min</code> (Phase1), potential energy defined as an integral; holds a.e., and a double-well witness proves the a.e. cannot be strengthened	Theorem
Landauer’s Principle	Derived from statistical mechanics (Phase3); the axiom-free reversibility core becomes a heat bound via the field <code>StatisticalMechanics.heat_eq</code>	Theorem + instance postulate
Symmetry Breaking implies Boundaries	Derived from pointwise potential minimization (Phase1); witnessed on a double well with a $\mathbb{Z}_2$ symmetry whose minimisation hypothesis is proved, not assumed	Theorem
Structural Resonance KL Bound	Gibbs’ inequality proved; the bound is the field <code>StructuralResonance.kl_bound</code> (Phase3_KLBound)	Theorem + instance postulate
Structural Resonance via Least Action	Gradient flow of σ in coupling space (Phase8), linked to <code>entropy_production_rate</code> ; finite substrates	Theorem
Phase-locking minimizes entropy production	For a <i>reciprocal</i> coupling kernel, the phase-locked field has the least entropy production of <i>any</i> field (<code>phase_locked_minimizes_entropy_of_symm</code> , Phase8 §2a). The mean-matching condition the earlier form assumed over the comparison class is derived, not imposed: the coupling term is anti-symmetric under exchanging its two sites, so total drift is conserved across phase fields (<code>total_drift_eq_of_symm</code>). Reciprocity is necessary — a one-way coupling on two sites strictly beats its own phase-locked state (Examples §16)	Theorem
Electrodynamic Discretization	Lebesgue-Bochner edge weights (Phase2) sitting on faces of the complex and partitioning the region’s stress-energy; mesh refinement converges, witnessed on $[0, 1)$. The scalar read off the stress-energy tensor is a marked modelling choice: the partition holds for any non-negative choice, and two instances differing only in it have different couplings (Examples §6)	Theorem + instance postulate

Conceptual Claim	Lean 4 Implementation	Status
Phase synchronization to Unity	Sheaf-theoretic gluing (Phase5) over a ThermodynamicCover , whose fields are now all local: the global section is constructed by probability_glue_unique and the former class field sync_to_section_eq is a theorem about it. Witnessed by two patches built from different mass profiles, each section constructed over its own patch, whose glued state is neither of them (Examples §14). The cover's equilibrium hypothesis is no longer assumed on every instance: ThermodynamicCover.ofConvergentTrajectory (Phase5.EquilibriumBridge) discharges it from kuramoto_tendsto_global_minimum , and Examples §17.1 is a three-patch cover whose phase field is the limit of a trajectory started <i>unsynchronized</i> . The overlap-agreement hypothesis is untouched	Theorem + instance postulate
Reflexive Topology (the Self)	Banach fixed point for a contracting self-prediction map (Phase6); metric, completeness and contraction are hypotheses, all three discharged on the three-site substrate (Examples §10), where global sections are proved to be exactly the finite measures on it, the metric is the uniform distance between their densities, and the self-prediction map contracts by exactly 1/2 without being constant. The avatar's link to the field is the predicate IsRestriction-Resonance , discharged on the same witness; under it a covering family of avatars determines the global state (eq_of_avatar_eq), and without it the Self theorem is unchanged (self_of_const-Resonance)	Theorem (witnessed)
Kuramoto Lyapunov descent	Descent chained to the minimum characterization by the rotating frame (Phase4.RotatingFrame); identical frequencies	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Kuramoto dynamics run	The initial value problem is well posed on all of $\mathbb{R}$: the field is globally Lipschitz and globally bounded (<code>kuramotoField.lipschitz</code> , <code>kuramotoField.norm_le</code>), and through every state at every instant passes exactly one trajectory, defined for all time (<code>kuramoto_cauchy_problem</code>). Along <i>every</i> trajectory of a zero-frequency system the dynamic potential converges (<code>dynamic.potential.tendsto</code>) and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite, equal to the potential's total drop (<code>dissipation.integral.tendsto</code>). The motion stops: Barbalat's lemma is proved here, Mathlib lacking it, and applied, so $\sum_i \dot{\theta}_i^2 \rightarrow 0$ along every trajectory (<code>velocity.sq.tendsto.zero</code>). Where it stops is proved under an initial-data condition: if the phases start within a quarter turn and the potential excess starts below $a/2$, where $a > 0$ bounds the coupling below, the trajectory converges to a phase-locked configuration that is a <i>global</i> minimiser, with $r^2 = 1$ (<code>kuramoto.tendsto.global.minimum</code>), by a Polyak–Łojasiewicz estimate rather than a LaSalle principle. Unconditionally this is false — splay and twisted states are equilibria — so the condition cannot be removed. Witnessed on three sites, where no closed form exists (<code>Examples §17</code>)	Theorem (conditional)
Continuous vs. Rigid Topology	Fixed wiring support at matched resource (<code>Phase7.Rigidity</code>), deriving the suboptimality Phase 7 assumed; typed continuum/discrete separation	Theorem
Critical Coupling Threshold ($K_c = 2D$)	Self-consistency equation (<code>Phase8.SelfConsistency</code>): at or below K_c , threshold included, only $r = 0$ solves it; above K_c <i>exactly one</i> $r > 0$ does (<code>supercritical.fixed.point.existsUnique</code>), so the non-negative solutions are $\{0\}$ below and $\{0, r\}$ above; that r tends to 0 as $K \downarrow K_c$ and increases strictly with K (<code>coherent.branch.strictMono</code>); a solution is a density reproducing its own order parameter. Conditional on the von Mises stationary density, which is an input, and on no link to a trajectory	Theorem (conditional)

109 2.1 Soundness of the formalization

110 Formalization is only as trustworthy as the axioms it rests on, and an axiom
111 that is inconsistent silently makes every theorem downstream of it vacuous

112 while leaving the build green. We report a case of exactly this, found by
 113 auditing our own development, because the failure mode is easy to reproduce
 114 and we have not seen it discussed in the applied-formalization literature.

115 Earlier drafts of this work declared five axioms. Three of them were
 116 individually inconsistent: each, on its own, proves **False**. All three shared
 117 one root cause — an axiom that constrains a symbol it does not itself bind.
 118 Concretely:

- 119 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
 120 energy functional as an *implicit* argument occurring only in its hypoth-
 121 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
 122 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields
 123 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
 124 this proves $1 = 0$.

- 125 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
 126 a bath-entropy difference *for every instance* of the class. An instance
 127 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.

- 128 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ
 129 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
 130 point mass against a uniform distribution refutes it.

131 The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`,
 132 had the same shape — global assertions pinning free fields of `LocalSectionSynchronization`

— and are refutable as soon as the probability presheaf admits two distinct global sections. (Both were demoted to class fields at the time; both have since been removed from the class altogether, for a separate reason given in Derivation 5.)

The correction is uniform, and we state it as a rule we now follow: *a physical postulate that mentions a class field must be a field of that class, never a standalone axiom quantified over all instances.* A postulate should be an obligation each model discharges, not a global claim about every model. Recast this way, the development declares no axioms at all. Two of the five became genuine theorems — pointwise vacuum minimization is now proved from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs’ inequality was already one — and the other three became instance obligations.

This makes non-vacuity the live question, since an uninhabitable class makes its theorems vacuous just as surely as an inconsistent axiom did. **Examples.lean** therefore constructs witnesses: a one-bit erasure system with a genuine bath that discharges **heat_eq** and reproduces $\Delta Q = k_B T \ln 2$; a perfectly resonant system discharging **kl_bound**; and a scalar field on a one-point spacetime discharging **ActionPrinciples**. Derivation 5’s gluing structures are witnessed too, which was until recently the largest open gap: **Examples.lean** builds a **ThermodynamicCover** over a three-site substrate carrying the discrete topology and its Borel σ -algebra, covered by two *overlapping* open patches whose intersection is a single shared site. The local sections are germ families of a phase-indexed finite measure $\mu_\theta =$

156 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the
 157 family is provably non-constant in θ (`phaseMeasure_not_const`), so the pe-
 158 riodicity obligation of the constructor `LocalSectionSynchronization.of-`
 159 `InvariantMeasure` is discharged by an argument about `cos` rather than satis-
 160 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`
 161 rather than assumed. `global_section_from_thermodynamics` then fires on
 162 this instance and glues the two patch-local sections into a unique global one.
 163 Two further covers answer the follow-up question of how non-degenerate such
 164 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 165 ing: the phase field can be made non-constant as a function, but not as a
 166 physical state, because `thermodynamic_equilibrium` forces every instance
 167 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 168 round, from two patches with independently chosen mass profiles that meet
 169 only on the site they share, and the section the gluing returns is neither of
 170 them. That is the same kind of result as `contracting_implies_const` in
 171 Derivation 6 — a proof that the witness is as strong as the class permits,
 172 rather than a witness that could be improved by trying harder.

173 What the witnesses establish is that the classes are inhabitable, so the
 174 theorems about them are not vacuous. What they do not establish is that
 175 a cortex is such an instance. In the cover above the phase field is already
 176 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 177 the equilibrium obligation dischargeable; getting a physical system into that
 178 configuration is the content of the informal dynamical argument, not of the

179 Lean development.

180 The same audit turned up a defect of a different kind, worth record-
181 ing because it is not an axiom problem and would not have been caught
182 by `#print axioms. mesh_refinement_convergence` — the claim that dis-
183 crete edge weights converge to the continuous energy as the mesh is refined
184 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
185 not merely unproved but false. Nothing required the edge regions to cover
186 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
187 all empty satisfied the mesh hypothesis while contributing nothing to the
188 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
189 quantified over a triangulation and then resolved the vertex type by instance
190 search rather than through the bound variable, so the variable was never
191 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
192 gence is now stated along a *sequence* of triangulations of a named region
193 S , each required to be regular — edge regions measurable, without self-
194 loops, disjoint across distinct edges, covering S , and anchored to the sim-
195 plicial data in the sense that a vertex lies in the closure of every region
196 of an edge at it. Closure rather than membership is forced: cells are dis-
197 joint, so a shared vertex belongs to at most one of them. The manuscript’s
198 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
199 and is therefore not symmetric, is shown to equal exactly the midpoint rule
200 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
201 tion, reading the double sum as a Riemann sum would be a category error.

202 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 203 the integrand, and convergence follows for uniformly continuous integrands
 204 as the mesh shrinks. The regularity conditions pull against each other, so
 205 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into
 206 half-open cells and checks that the approximation genuinely moves $-\frac{1}{2}$ at
 207 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 208 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 209 the accompanying simulation measures; that would need a midpoint error
 210 term with a second derivative, which we have not developed.

211 A third defect was one of bookkeeping across files, and it is the kind that
 212 survives any number of green builds because each half is individually cor-
 213 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 214 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 215 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 216 — phase-locked in both directions — and everything downstream of it, in-
 217 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 218 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 219 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 220 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 221 it has no minimum for phase-locking to attain, and the sentence “the phase-
 222 locked state minimizes the Lyapunov potential” was, read literally, false of
 223 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 224 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical

225 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 226 frequency system — which is exactly the system Derivation 5 already worked
 227 with — the two potentials coincide there, and the descent result transports
 228 onto the coupling potential whose minimizers are the phase-locked states.
 229 Phase differences, phase-locking and the magnitude of the order parameter
 230 are all proved invariant under the change of frame, so nothing observable
 231 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 232 two oscillators with a common frequency, discharging every hypothesis. The
 233 restriction is real and we state it plainly: the reduction is exact only for
 234 identical frequencies. A genuine spread of natural frequencies leaves residual
 235 detunings in the reduced system, phase-locking then requires coupling above
 236 a critical K_c , and K_c is nowhere in our Lean development — it is studied
 237 numerically, as Table 1 records.

238 A fourth defect was dimensional, and it concerned the one row of Table 1
 239 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 240 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 241 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 242 against an unnormalized measure carries the substrate’s mass twice over,
 243 so a constant kernel K on a substrate of total mass m contributed Km^2 .
 244 The comparison therefore depended on how large the substrate was rather
 245 than on how strongly it was coupled, and any kernel whatever could be
 246 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
 247 ure concretely: on a two-site substrate carrying counting measure, a sys-

248 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
 249 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
 250 satisfied the old predicate. The fix is to require the substrate’s measure to
 251 be a probability measure, which is the normalization in which the mean-
 252 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 253 then gives that a constant kernel has mean-field strength exactly K , and
 254 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 255 We flag what this does and does not do. It makes the predicate say something
 256 whose truth depends only on the physics, and §9 witnesses it on both sides of
 257 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 258 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 259 at $K = 2D$, which is a separate question from whether the predicate is di-
 260 mensionally sound. That bifurcation is proved below, in both directions,
 261 from an assumed stationary density; deriving the density itself needs the
 262 Fokker–Planck operator and is not done.

263 We draw no grand moral from this beyond the obvious one. The value
 264 of a proof assistant here was not that it certified our argument, but that it
 265 made a specific, checkable claim about what had been assumed — and that
 266 claim turned out to be false in a way no amount of reading the prose would
 267 have revealed.

268 3 Axiom 1: Symmetries and Phase Space

269 Our starting premise relies on fundamental geometry. In our Lean formaliza-
 270 tion (`Phase1_Primitives.lean`), continuous spacetime symmetries (e.g., the
 271 Poincaré group) act upon field states, and the invariance of the system’s total
 272 energy under that action is recorded by the class `SymmetryInvariantAction`.
 273 Physically, it is such an invariance that Noether’s theorem converts into the
 274 conservation of energy and momentum. The scope of the *formal* claim is nar-
 275 rower and we state it plainly: the invariance is a class field, i.e. an assumption
 276 a model must discharge, and no conserved quantity is ever constructed from
 277 it. There is no Noether theorem in our Lean, and nothing downstream of this
 278 section depends on one. The class is now inhabited — `Examples.lean` §11
 279 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
 280 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
 281 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
 282 changed by the symmetry, so an energy-minimising field and its image have
 283 equal energy and differ. That is symmetry breaking exhibited rather than
 284 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 285 no one-parameter family to differentiate along in the first place. Simultane-
 286 ously, discrete permutation symmetries dictate the statistical mechanics of
 287 any physical system, defining a strictly finite volume of phase space.¹

¹While the Poincaré group and discrete permutations serve as accessible physical ex-
 amples, the underlying Lean formalization does not hardcode specific physics. It defines
 the system at a more primitive mathematical level, relying entirely on generic Pseudo-
 Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-

288 Because phase space is finite, any localized physical system possesses a
 289 mathematical limit on the amount of information (distinguishable states) it
 290 can embody at any given moment.

291 4 Derivation 1: Symmetry Breaking and the 292 Inevitability of Boundaries

293 A perfectly symmetric universe is the uniform vacuum; no distinct entities
 294 exist. For any dynamic process to occur, the continuous symmetries of the
 295 background field must undergo spontaneous symmetry breaking.

296 When a field drops to a lower, asymmetric energy state, topology dic-
 297 tates the natural creation of defects (such as domain walls or monopoles).
 298 This constraint necessitates the creation of a physical boundary—a localized
 299 topological distinction between an “inside” and an “outside.” While such
 300 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
 301 entity, it is not sufficient for agency. The boundary is relational, defined by
 302 the twisted geometry of the field surrounding it, and must be coupled with
 303 non-equilibrium thermodynamics to exhibit active behaviors.

ics). What the formalization supplies is the vocabulary in which symmetries, conserved quantities and phase-space constraints can be *stated*; it does not derive them. All three of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its invariance in §11, the real line with the Euclidean metric in §12 — but the instances are witnesses of inhabitability, not of physics: no theorem in the development consumes a symmetry group or a metric, so this section still contributes vocabulary rather than content.

304 5 Derivation 2: The Thermodynamic Con- 305 straint

306 This newly formed topological boundary is subjected to gauge perturbations
307 from the “outside” environment. Because the internal phase space is finite
308 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
309 and must overwrite old internal states.

310 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
311 formation irreversibly decreases internal entropy, mandating that the sys-
312 tem exhaust heat into the environment. Philosophers of physics (such as
313 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
314 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure
315 (a many-to-one mapping in abstract computational states) with *thermody-*
316 *namic* erasure (a many-to-one mapping in physical phase space). They argue
317 that logical erasure could conceptually be implemented by an injective phys-
318 ical process without dissipating heat.

319 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
320 addresses this critique by defining erasure as a non-injective mapping on the
321 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
322 tion of the bipartite universe (System + Bath) as strictly injective, repre-
323 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
324 to abstract logical erasure, but is a mandatory mathematical consequence of
325 any local physical state space contraction under globally reversible dynamics.

326 If a cognitive system’s internal topology forces a contraction of its physical
 327 state space to accommodate structural resonance, heat dissipation becomes a
 328 thermodynamic requirement. A stable boundary thus constitutes a dissipa-
 329 tive structure (Prigogine & Stengers, 1984), operating as a non-equilibrium
 330 steady state that consumes free energy to pay the thermodynamic cost of
 331 erasure. Continuous thermal fluctuations play a constructive role in estab-
 332 lishing boundaries between states while contributing to the system’s flexible,
 333 adaptive response.

334 **6 Derivation 3: Structural Resonance as a** 335 **Primitive**

336 In physical systems, time evolution follows a deterministic trajectory that
 337 minimizes thermodynamic entropy production. To reach this minimal dissi-
 338 pation state, the internal geometry of the structure must physically deform.

339 If we model the external environment as a stochastic driving force $P(ext)$
 340 and the system’s internal state transitions as $Q(int)$, entropy production σ is
 341 strictly bounded from below by the Kullback-Leibler divergence $D_{KL}(P||Q)$.
 342 Gradient descent on σ necessarily drives $D_{KL}(P||Q) \rightarrow 0$, physically forcing
 343 the internal transition probabilities to match the statistical symmetries of
 344 incoming perturbations. We define this physical phenomenon as *structural*
 345 *resonance*.

346 This system does not possess a teleological drive to “survive”; rather,

347 structurally mirroring the outside world is simply the lowest-energy attrac-
348 tor state. This provides a thermodynamic mechanism for both the General
349 Resonance Theory of Consciousness (Hunt & Schooler, 2019) and the Free
350 Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy, 2013; Clark,
351 2016; Spisák & Friston, 2026). What neuroscience observes as “prediction er-
352 ror minimization” is a physical system undergoing gradient descent to avoid
353 dissipating heat against an unpredictable environment.

354 This necessity also addresses the symbol grounding problem (Harnad,
355 1990). Internal states are physically indexed to the environment by the ener-
356 getic imperative to avoid heat dissipation, rather than through arbitrary syn-
357 tactic assignment. Manipulating abstract variables or counterfactuals forces
358 the system to maintain internal states that deviate from immediate sensory
359 resonance, explaining the continuous injections of free energy required for
360 abstract symbolic reasoning.

361 **7 Derivation 4: Macroscopic Scaling and Elec-** 362 **trodynamics**

363 To resonate with highly complex environments, the system must scale up,
364 coupling millions of minimal dissipative structures. A highly coupled network
365 cannot allow all of its nodes to reach their lowest energy states simultane-
366 ously. This creates *geometric frustration*, resulting in a jagged phase space
367 of near-equal energy states that provides the physical capacity for complex

368 memory.

369 To manage this frustrated phase space without excessive dissipation, the
370 system undergoes coarse-graining (governed by the Renormalization Group).
371 At the macroscopic scale of biology, emerging paradigms in neuroscience sug-
372 gest these topological constraints manifest as classical continuous electromag-
373 netic fields. Building on seminal EM field theories of consciousness (McFad-
374 den, 2002; Pockett, 2012) and recent work demonstrating that macroscopic
375 electric fields (ephaptic coupling) govern representational drift, form memory
376 networks, and sculpt neural infrastructure (Pinotsis & Miller, 2022, 2023a,b),
377 we posit that *the continuous macroscopic EM field is the primary dissipative*
378 *structure interacting with the universe*. Instead of discrete wires, neurons in-
379 duce fields that shift the membrane potentials of neighboring cells. Modeling
380 and *in vitro* experiments confirm these endogenous fields are strong enough
381 to guide network activity and contribute to brain complexity (Goldwyn &
382 Rinzel, 2016; Ruffini & Santarnecchi, 2020; Cunha & Lima, 2024; Fröhlich &
383 McCormick, 2010). As proposed by the CEMI field theory, the continuous
384 electromagnetic field enables spatial integration of information, providing a
385 physical medium for a unified conscious gestalt (McFadden, 2020).

386 This continuous medium is physically supported by astrocytic syncytia—
387 non-neuronal networks joined by gap junctions that provide macroscopic cy-
388 toplasmic continuity for energy and ion flows (Verkhratsky & Nedergaard,
389 2018). Managing the thermodynamic cost of this state requires sophisticated
390 micro-bioenergetics. Dynamic mitochondrial networks function as deep-level

391 Mitochondrial Information Processing Systems (MIPS), transducing biolog-
392 ical and environmental information to sustain the coherence of the macro-
393 scopic field (Picard & Shirihai, 2022). Discrete synaptic firings, coupled
394 with mitochondrial micro-mechanics, generate and sustain the field. Because
395 the external environment is continuous, achieving maximal thermodynamic
396 structural resonance is optimally realized within this continuous macroscopic
397 medium (van Bree et al., 2025).

398 8 Derivation 5: Global Sections and the Unity 399 of Consciousness

400 The system is now characterized by a continuous macroscopic electrodynamic
401 field generated by millions of frustrated micro-components, forced to resonate
402 with the environment.

403 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-
404 tors guarantees that, past a critical coupling threshold, these distributed
405 local fields undergo a phase transition into unified synchronization (Bauer et
406 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
407 phase-locked state minimizes the Lyapunov potential of the system, in the
408 precise sense discussed in Section 2.1: the functional minimized is the cou-
409 pling potential read in the frame rotating with the natural frequency, which
410 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
411 system of identical oscillators.

412 This phase transition mathematically glues local synchronized states into
 413 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
 414 The EM field acts as a single, indivisible topological space, providing the
 415 *Unity of Consciousness*—a necessary condition for integrated experience that
 416 solves the binding problem. While Global Workspace Theory (Baars, 1997;
 417 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 418 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 419 our derivation provides a continuous physical mechanism. The contrast is
 420 worth drawing carefully, because both theories are addressing the same ex-
 421 planandum from different directions.

422 We state the scope of the Lean result precisely, because it is narrower
 423 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 424 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 425 `phase_locked` characterizes those minimizers exactly: they are the phase-
 426 locked configurations, in both directions. Every instance is therefore already
 427 phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single
 428 condition on its local data — that patches at a common phase agree where
 429 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 430 sections compatible. `probability_glue_unique`, which is the sheaf property
 431 of the probability presheaf and contains no physics at all, then glues that
 432 family into exactly one global section.

433 An earlier version of this structure proved less than the prose claimed,
 434 and we record the correction because it changed what the theorem says

435 rather than how it is stated. `LocalSectionSynchronization` used to carry
 436 a *global* section as a class field, together with a second field declaring every
 437 local section to be a restriction of it. The global object therefore existed
 438 before the sheaf condition was ever consulted, and `global_section_from_`
 439 `thermodynamics` could do no more than rule out competitors to an object
 440 the instance had supplied. Both fields are gone. Every field of the class
 441 now mentions a single patch or a single overlap; the global section is con-
 442 structed (`ThermodynamicCover.invariantMeasure`); and the removed field
 443 survives as a *theorem* about that construction (`ThermodynamicCover.sync_`
 444 `to_section_eq`), so the local sections are restrictions of one common measure
 445 because the sheaf condition says so, not because an instance was required to
 446 name one.

447 `Examples.lean` §14 is the witness that the difference is not notational.
 448 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
 449 the three sites, agreeing only at the site they share, where both put mass 1
 450 — which is what discharges the overlap condition, by computing two num-
 451 bers rather than by construction. The section the theorem returns has pro-
 452 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
 453 `rightW`), and neither of them restricts correctly to both patches (`leftW_`
 454 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 455 tion that no single patch had. `Examples.lean` §13, built the older way
 456 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 457 is retained alongside it for the negative result it carries: its phase field is not

the constant function — the second patch a full turn ahead of the first — and by `ThermodynamicCover.phase_locked` that is as far from constant as any instance can get, since no instance can carry phases differing by anything a measure is allowed to see.

Two hypotheses carry physical content, and until recently each was an obligation on every instance. One still is: that synchronized patches agree on their overlaps is a modelling assumption, not a consequence of any dynamics we formalize. The other, `thermodynamic_equilibrium`, asks that the cover already sit at the potential minimum, and no theorem says an *arbitrary* system’s dynamics takes it there. “A thermodynamic phase transition into unity” is therefore still doing causal work that no general Lean result supports, though it is no longer entirely without one: `Examples.lean §15` exhibits a two-oscillator trajectory that starts a quarter turn out of phase and converges to the global minimum of the dynamic potential, with the order parameter tending to 1. That trajectory is not an isolated construction: `kuramoto_cauchy_problem` proves the Kuramoto initial value problem well posed on all of $\mathbb{R}$, so through every initial state there passes exactly one trajectory, and the exhibited one is the only solution through its own. Nor is its settling down peculiar to it: along every trajectory of a zero-frequency system the dynamic potential converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite and equal to the potential’s total drop (`dynamic_potential_tendsto`, `dissipation_integral_tendsto`). The motion also stops: a finite integral does not by itself force its integrand to zero, but the dissipation rate is

481 Lipschitz in time, so Barbalat's lemma applies, and every trajectory satis-
 482 fies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_zero`; Mathlib has no Barbalat, so it
 483 is proved here as `tendsto_zero_of_lipschitz_of_integral_le`). That last
 484 step — locating where the motion stops — is now taken, though condition-
 485 ally, and by a different route than we had recorded. We had marked it as
 486 blocked on the LaSalle invariance principle, which Mathlib does not have,
 487 and on the absence of a compact invariant set in the state space $V \rightarrow \mathbb{R}$.
 488 That was the wrong diagnosis. LaSalle is the tool for a system with a de-
 489 creasing functional and nothing else; this one has more. The flow is a gra-
 490 dient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an identity rather
 491 than an inequality; the minimisers of V are characterised exactly; and on
 492 the region where every pair of phases lies within a quarter turn a Polyak-
 493 Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is the poten-
 494 tial's excess above its minimum and $a > 0$ is a lower bound for the coupling
 495 (`lojasiewicz_estimate`). Its proof is two extremal indices and the iden-
 496 tity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase is
 497 pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 498 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 499 as well, because the invariant region is produced by the Lyapunov function
 500 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 501 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 502 continuity on the connected half-line then prevents any pair from chang-
 503 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to

504 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 505 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 506 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 507 `minimum`: if the phases start within a quarter turn of each other and the initial
 508 excess is below $a/2$, the trajectory converges to a phase-locked configuration
 509 which is a *global* minimiser of the dynamic potential, with order parameter
 510 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 511 reached rather than assumed. `Examples.lean` §17 fires it on three sites with
 512 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 513 tion: the trajectory comes from the existence theorem and its limit from the
 514 convergence theorem, so neither is written down.

515 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
 516 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 517 cover whose phase field is the limit of such a trajectory and returns an in-
 518 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 519 *vergence*; the cover's coupling matrix is the coupling matrix of the dynamics
 520 rather than a separately chosen one, so the system whose equilibrium the
 521 class asserts and the system whose trajectory is run are the same system.
 522 `Examples.lean` §17.1 is the resulting witness: three patches over the three-
 523 site substrate, each carrying its own mass profile on its own open set, above a
 524 phase field that no line of the construction writes down. The trajectory that
 525 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
 526 `start_not_locked`), so on this cover the configuration Derivation 5 requires

527 is arrived at rather than posited. The state the three patches glue to has
 528 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
 529 `ne_glued`), so one cover now carries both the emergence reading of the gluing
 530 and the derived reading of the equilibrium. What is still assumed on it is the
 531 overlap agreement, and nothing else.

532 The two hypotheses cannot be dropped, and this is the honest scope of
 533 the result. Splay and twisted configurations are equilibria of the same flow,
 534 so “every trajectory reaches the unified state” is false as a general statement,
 535 and the initial-data condition is the standard arc condition under which the
 536 true statement holds. What is claimed is therefore conditional — unity is
 537 reached from configurations already sufficiently coherent — and not that an
 538 arbitrary initial state is carried into unity. What Lean establishes about the
 539 gluing itself is the other half: *given* locally synchronized patches, a global
 540 state exists, is unique, and is produced by the gluing rather than presupposed
 541 by it.

542 Integrated Information Theory begins from phenomenological axioms and
 543 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
 544 with the exclusion postulate selecting a single maximum- Φ complex as the
 545 locus of experience. Our sheaf-theoretic global section plays a structurally
 546 analogous role — it too picks out a distinguished unified object — but it
 547 is arrived at from the opposite end: not postulated from the character of
 548 experience, but forced by a thermodynamic phase transition in a physical
 549 field. This buys two things IIT does not have. First, the unity condition

550 is checkable from field dynamics rather than requiring the computation of
 551 Φ , which is intractable for systems of realistic size and whose definition has
 552 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
 553 out rather than being postulated: gluing either succeeds on a given cover or
 554 it does not, so there is no separate principle needed to rule out overlapping
 555 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
 556 that Φ can be high in systems with no plausible claim to experience, such
 557 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
 558 to the mirror objection, since a physical system could in principle satisfy
 559 the gluing condition without being cognitively interesting. We take this
 560 seriously rather than dismissing it: our reply is that gluing is necessary but
 561 not sufficient, and that the topological boundary of Derivation 1 and the
 562 reflexive structure of Derivation 6 are additional required conditions. IIT, by
 563 contrast, treats Φ as both necessary and sufficient.

564 Global Workspace Theory faces a complementary difficulty. It specifies
 565 a functional architecture — a limited-capacity workspace broadcasting to
 566 specialized modules — without specifying what physical property of broad-
 567 casting makes the broadcast content conscious, leaving the theory open to the
 568 charge that it explains access consciousness while leaving phenomenal con-
 569 sciousness untouched (Block, 1995). Our framework can be read as supplying
 570 the missing physical substrate for the workspace: the continuous macroscopic
 571 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
 572 old gives a physical condition for ignition rather than a functional stipulation

573 of it. Where GWT is agnostic about implementation, we make a falsifiable
 574 commitment — unity requires a continuous physical medium, which predicts
 575 that a functionally equivalent discrete broadcast architecture would not be
 576 conscious. We regard this as a genuine cost of the position, not a free ad-
 577 vantage: it is the point at which our framework departs most sharply from
 578 computational functionalism and where it is most readily falsified. Contrast
 579 this derivation with panpsychism: consciousness is not assumed as a funda-
 580 mental property of matter. It requires the initial existence of a topological
 581 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 582 dynamic phase transition into unity that occurs when integrated spatial cou-
 583 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 584 the foundational topological boundary and the subsequent mathematically
 585 defined phase transition, the system remains a fragmented, non-conscious
 586 collection of micro-states. A unified field of perception, however, is not yet
 587 a “Self.”

588 **9 Derivation 6: Reflexive Topology and the** 589 **Emergence of the Self**

590 While a global section provides Unity, a “Self” requires reflexivity. The
 591 unified EM field is a dissipative structure that must resonate with external
 592 perturbations while managing its own internal phase transitions. Time evo-
 593 lution naturally drives the field to minimize thermodynamic friction with its

594 own internal state-changes.

595 This mandates *auto-resonance*: the field must predict itself. The “Self”
596 emerges as a higher-order topological feature when the internal geometry
597 folds to include a low-dimensional avatar of its own computational and causal
598 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)
599 (Levin, 2019). The system is unified, predictive, and explicitly maps its
600 own distinct existence from the environment. This architecture provides the
601 physical foundation for a self-aware consciousness, bounded by the continuous
602 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).
603 We are explicit about what carries the weight here. `reflexive_topology_implies_self`
604 is the Banach fixed-point theorem with the physical commitments in its hy-
605 potheses: the space of global sections is assumed nonempty, metric and com-
606 plete, and the predictive map is assumed to contract by $1/2$. `Examples.lean` §10
607 discharges all four on the three-site substrate of Derivation 5, and the metric
608 is no longer a placeholder. The section first pierces the sheafification: on
609 a discrete substrate every point has a smallest open neighbourhood, so the
610 mass a germ carries at a point is well defined, and `massEquivOn` proves that
611 over *any* open the sections of the probability sheaf *are* the finite measures
612 on that open, read as densities against counting measure — the global case
613 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
614 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
615 correspondence had to be constructed. The metric on sections is then the
616 uniform distance between the densities of the measures they glue to — equiv-

617 alent to their total variation on a finite substrate, though not equal to it —
 618 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 619 of densities, not by the degenerate route of showing Cauchy sequences are
 620 eventually constant. The self-prediction map is the relaxation “average the
 621 current field with the baseline,” and we prove it contracts distances by *ex-*
 622 *actly* one half, that it is not constant, and that its fixed point is unique. An
 623 earlier version of this witness used the 0/1 metric, under which every con-
 624 traction is constant; that theorem is kept in the source as the record of why
 625 the earlier witness was empty. A separate gap, closed at the same time, con-
 626 cerned the avatar rather than the fixed point. `ReflexiveBoundary` carries a
 627 field `auto_resonance` sending the global state to a section over the avatar re-
 628 gion, and nothing constrained it: a constant function was a legal avatar, and
 629 `reflexive_topology_implies_self` consumes only the predictive model, so
 630 no theorem in the development mentioned the field the avatar is supposed to
 631 be reading. The constraint is now stated as a predicate rather than added
 632 to the structure — `IsRestrictionResonance`, that the avatar’s state is the
 633 presheaf restriction of the global state — and two theorems say why it is
 634 the right one. In the positive direction, if a family of resonant avatars *cov-*
 635 *ers* the substrate then the global state is determined by their local readings
 636 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 637 recovered from them, by the same sheaf condition that Derivation 5 uses.
 638 In the negative direction, replacing any avatar by a constant leaves a legal
 639 `ReflexiveBoundary` with the same region, the same predictive model and

640 the same Self (`self_of_constResonance`), so the fixed-point theorem cannot
 641 see the difference and the predicate is doing the work rather than restating
 642 it. Both halves are witnessed on the three-site substrate, where the avatar
 643 region provably separates two states of the field, so the condition is a con-
 644 straint there and not a formality. What is still not established is that any field
 645 dynamics produces this particular map or performs the restriction, or that
 646 the contraction constant $1/2$ is read off an entropy production rate rather
 647 than built into the map's definition. We also note a negative result found
 648 in the course of this work: the Lévy–Prokhorov metric, which the natural
 649 weak-convergence reading would suggest, does *not* support this argument on
 650 a discrete substrate, because the ε -thickening of a set is the set itself for $\varepsilon < 1$
 651 and the relaxation map is consequently not a Lévy–Prokhorov contraction.

652 **10 Derivation 7: Continuous Stochastic Ther-** 653 **modynamics and Field Theory**

654 To move beyond finite graph analogies, we model the macroscopic electro-
 655 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
 656 continuous spatial phenomenon, not a discrete network. We extend the
 657 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
 658 (`Phase8_ContinuousField.lean`).

659 Subject to thermodynamic noise D , the system undergoes non-equilibrium
 660 stochastic thermodynamics. Rather than a static Landauer bound, the field

661 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (1)$$

662 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
 663 dergo gradient descent on this entropy production functional.

664 That gradient descent is a theorem rather than a modelling assertion,
 665 for finite substrates. Reading σ as a functional of the coupling kernel at
 666 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
 667 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (2)$$

668 and `sigmaOfKernel_eq_entropy_production_rate` identifies the functional
 669 being differentiated with `entropy_production_rate` itself whenever the sub-
 670 strate carries counting measure. `structural_resonance_decreases_entropy_production`
 671 then concludes that σ is non-increasing along any coupling trajectory relax-
 672 ing down its own gradient at a positive rate c . Earlier drafts proved this
 673 only for an abstract functional S on an abstract inner-product space with
 674 the gradient supplied as data; the identification of that S with σ was infor-
 675 mal, and Table 1 flagged the row accordingly. Two limitations stay explicit,
 676 and the first is narrower than we previously reported. The derivation is for
 677 *finite* substrates. We had recorded the obstacle to the continuum case as

678 differentiation under the integral sign with respect to the kernel; that was
 679 a misdiagnosis. σ is not a general functional that happens to be written as
 680 an integral — it is a quadratic functional of an *affine* image of the kernel,
 681 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
 682 constant times the squared norm of the drift. A squared norm is Fréchet
 683 differentiable on any real inner-product space and an affine map is its own
 684 derivative, so the composite differentiates by the chain rule with no limiting
 685 argument. `hasFDerivAt_quadratic_of_affine` states that in full generality,
 686 and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives the finite gradient
 687 from it, showing that the closed form displayed above is a convenience rather
 688 than the content. What the continuum case still needs is therefore one object:
 689 the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded operator
 690 $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral ker-
 691 nel is bounded by 1, so on a finite measure space the operator norm is at most
 692 $\mu(M)^{1/2}$. We have not built it, because it requires constructing a continu-
 693 ous linear map between L^p spaces and we have not done that bookkeeping;
 694 we record the corrected scope because “construct one bounded operator”
 695 and “differentiate under the integral sign” are different sizes of task. The
 696 second limitation is unchanged: the phase field is held fixed throughout, so
 697 this is plasticity of the coupling at frozen phases, not joint (θ, K) dynamics.
 698 **Examples.lean** §5 exercises the whole chain on a two-site substrate whose
 699 coupling relaxes as e^{-t} and whose entropy production decays as e^{-2t} , con-
 700 firming that the gradient is not identically zero and that the descent is strict

701 rather than vacuously monotone.

702 A second result about σ concerns where the descent is headed, and it has
703 been strengthened since the last draft in a way that also corrects an estimate
704 we had recorded. That the phase-locked field minimizes σ was proved only
705 against competitors sharing its spatial mean drift — a restriction that left
706 the theorem close to Jensen’s inequality and no more — and we had judged
707 that lifting it would require modelling how the mean drift varies with the
708 competitor. That judgement was wrong. For a *reciprocal* kernel, one with
709 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand
710 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because
711 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
712 its own negative and the total coupling contribution vanishes. The total drift
713 $\int(\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`
714 `symm`): coupling moves drift between sites but neither creates nor destroys it.
715 Evaluating that conserved quantity at the phase-locked configuration, where
716 the drift is constant, pins its value, and the mean-matching hypothesis follows
717 for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
718 restricted statement. Reciprocity is the physically intended condition — an
719 ephaptic field acts on both sites of a pair alike, and it is the same symmetry
720 `ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
721 not a formality, and we prove that rather than assert it. `Examples.lean` §16
722 exhibits a *one-way* coupling on two sites, in which a feels b and b does not
723 feel a , whose phase-locked field is strictly beaten: displacing the second phase

724 to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
725 ond, halving σ . So “phase-locking minimizes entropy production” is false for
726 general kernels and true for reciprocal ones, and the hypothesis cannot be
727 removed by working harder.

728 We define the critical coupling threshold $K_c = 2D$. The phase transition
729 to the unified “Global Section” occurs when the mean-field spatial coupling
730 exceeds this thermodynamic noise threshold, linking the emergence of Unity
731 to physical temperature and noise.

732 Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by
733 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that
734 is ordinary real analysis. For the noisy mean-field Kuramoto model with
735 identical frequencies, the stationary density at order parameter r is the
736 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$
737 $R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
738 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
739 tegration by parts with $u = \sin$, $v = -e^{a\cos}$ gives an exact identity with
740 no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a times
741 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
742 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
743 tor of two is the sharper bound $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
744 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
745 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
746 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor

747 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
 748 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 749 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 750 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 751 ther facts about the threshold have since been proved, and they narrow the
 752 gap between this row and a full “Theorem” to a single named statement.
 753 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`
 754 `pos`, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the
 755 self-consistency map is tangent to the diagonal at the origin but falls strictly
 756 below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 757 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 758 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 759 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coher-
 760 ent branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third,
 761 and using only $E < 1/2$ together with compactness, every coherent solution
 762 tends to zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the
 763 branch emerges continuously from the incoherent state rather than appearing
 764 at a finite distance from it, which is what makes the transition supercritical
 765 in the technical sense. That statement quantifies over *every* solution, so it
 766 does not presuppose the uniqueness it does not have. The level-set form
 767 also says exactly what uniqueness is: the injectivity of E on $(0, \infty)$. That
 768 is now proved, in the stronger form that E is strictly decreasing on $[0, \infty)$
 769 (`vonMisesSRatio_strictAntiOn`), and by an argument that avoids the route

770 we had expected to need. Differentiating the exponential family in its natu-
 771 ral parameter gives $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, which costs differentiation
 772 under the integral sign and leaves a covariance that is not sign-definite point-
 773 wise: symmetrising with an independent copy gives $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$
 774 with $X = \cos \theta$, an integrand changing sign with $X + X'$. Nothing in the
 775 proof differentiates anything. The target is kept as the two-point inequality
 776 $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: the
 777 two reflections that proved $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing
 778 $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$, after which
 779 $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both
 780 being governed by $\cos x$ against $\cos y$. This is the symmetrisation $X \mapsto -X$
 781 that the covariance form needed the tilt to perform, done once at the start; its
 782 analytic content is one identity, $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$
 783 with $\cosh$ increasing in $|\cdot|$. *Then cross*: instead of integrating that inequality
 784 over a square, let α be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a
 785 point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem; eval-
 786 uating at $y = x_0$ and integrating in x alone gives the result, since the term
 787 carrying $W(b, x_0)$ vanishes by the choice of α . No product measure and no
 788 Fubini appear. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no
 789 fold — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 790 and gives **besselRatio_strictMono**: R is strictly increasing on all of $\mathbb{R}$, the
 791 monotonicity of the Bessel ratio our earlier drafts recorded as unproved.

792 The converse half is **supercritical_fixed_point_exists**: for $K > K_c$

793 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 794 this to require a second-order expansion of the Bessel ratio, and it does
 795 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 796 tion the integration by parts already supplies — the only facts needed are
 797 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 798 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 799 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 800 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 801 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
 802 The intermediate value theorem closes the gap. Continuity of E is Math-
 803 lib's continuity of a parametric interval integral with jointly continuous in-
 804 tegrand, so no dominated-convergence argument is written by hand, and
 805 the existence proof never uses monotonicity of R : the fixed point is pro-
 806 duced by the intermediate value theorem, not by iterating the map. That
 807 R is in fact strictly increasing is proved below, and no threshold result de-
 808 pends on it. `critical_coupling_is_threshold` packages the two halves,
 809 and `exhibits_phase_transition_coherent` applies the supercritical half to
 810 any substrate satisfying the predicate of the previous paragraph — so ex-
 811 ceeding K_c now implies the existence of a coherent stationary order pa-
 812 rameter rather than merely comparing against a named number. With
 813 uniqueness available both are sharpened — `supercritical_fixed_point_`
 814 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 815 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`

816 `transition_unique_coherent` — and the two monotonicities together fix
817 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
818 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
819 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
820 order parameter grows strictly with the coupling, which with the continuity
821 at threshold is the qualitative bifurcation diagram.

822 One more step makes the equation mean in Lean what it means on paper.
823 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
824 r with a suggestive name, and nothing in the source says it has anything
825 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the
826 identification. `vonMisesDensity` is the normalized weight, proved to be a
827 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
828 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
829 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
830 actly $R(a)$ — real, because the density is even, so the mean direction van-
831 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
832 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
833 it induces has order parameter r . Read through this, the existence theorem
834 says that above K_c there is a stationary density whose own order parameter
835 is positive, and the solution $r = 0$ is identified with the uniform density,
836 whose order parameter is zero.

837 What remains not established, and we separate it carefully. The sta-
838 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$

839 $(\omega + K \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
 840 and uniqueness of its stationary solution, and a spectral stability argument,
 841 none of which Mathlib has. Of the coherent branch we prove existence,
 842 uniqueness, continuous emergence at the threshold and strict growth in K ,
 843 but not that it is the dynamically selected one: nothing in that file men-
 844 tions a trajectory, so stability and selection are not statements it can make.
 845 And while `selfConsistency` is now connected to the order parameter of
 846 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 847 integral, `order_parameter_complex` is an average over finitely many oscilla-
 848 tors, and relating them is the mean-field limit — propagation of chaos for
 849 the finite Kuramoto system — which is a research programme rather than
 850 a lemma. That single gap is now the whole of what separates our threshold
 851 predicate from a statement about dynamics. Table 1 records the row ac-
 852 cordingly as a theorem about the self-consistency equation from an assumed
 853 density, not as a derivation of the threshold from the dynamics. Perceptual
 854 anomalies such as visual snow can be modeled as boundary conditions where
 855 microscopic thermal noise (D) fails to be fully coarse-grained away by the
 856 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

857 Clinical evidence for this phase transition is observed during general anes-
 858 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
 859 that anesthetics like propofol induce unconsciousness not by suppressing neu-
 860 ral firing, but by disrupting the phase alignment of high-frequency cortical
 861 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-

862 tural resonance of the continuous field, forcing the integrated spatial coupling
863 below the critical threshold K_c and fracturing the unified Global Section back
864 into disjoint, unconscious local fields.

865 **11 The Sleep-Inertia Timescale Mismatch**

866 Most of what this framework asserts is shared with some rival: that corti-
867 cal synchrony matters, that dissipation constrains cognition, that unity is a
868 physical rather than a computational property. One prediction is not shared,
869 and it follows from the framework’s specific commitment — that the medium
870 carrying the unified state is a continuous field whose effective coupling de-
871 pends on the physical geometry of the extracellular space, not on the rhythms
872 alone. Where that commitment differs from a purely electrical account, it
873 differs in an observable way, and we set out the difference precisely enough
874 to be wrong.

875 The daily loss of consciousness during sleep is modeled here as an active
876 physical disruption of the medium rather than a quiescence of the oscillators.
877 During sleep, astrocytic shrinkage expands the extracellular space, allowing
878 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste
879 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
880 itself, disrupting the continuous structural resonance and preventing the co-
881 herent sheaf-theoretic gluing that conscious unity requires on this account.
882 The oscillators need not stop for the field to fragment; the geometry they are

883 coupled through is what changes.

884 The prediction concerns sudden awakenings. When jolted awake, the lo-
885 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
886 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
887 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
888 cess CSF out of the interstitial pathways operates on a timescale of seconds
889 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
890 slower. During that lag the electrical oscillators are synchronizing across a
891 waterlogged topology, and the fragmented structural resonance is what is ex-
892 perienceed as grogginess. The unified Global Section re-establishes only when
893 astrocytic volume finishes matching the state of the electrical rhythms.

894 **Why this implies a delayed sigmoid.** The two timescales separate
895 cleanly, and that separation is what makes the prediction sharp rather than
896 merely qualitative. Write $K(t)$ for the effective coupling of the continuous
897 field, which on this account is a monotone function of the reconstricting
898 extracellular geometry, and let $r(t)$ be the order parameter of the phase
899 field. The electrical variable is fast: at every instant the oscillators are al-
900 ready at the configuration appropriate to the current K , so r tracks the
901 equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
902 able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
903 seconds. What Derivation 7 establishes about that branch is exactly what
904 the functional form needs, and each of the three pieces is a theorem rather

905 than a picture: at or below the threshold the incoherent solution $r = 0$ is the
 906 *only* non-negative solution of the self-consistency equation (`fixed_point_eq_`
 907 `zero_of_le_critical`); above it the coherent solution exists and is unique
 908 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at
 909 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
 910 and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
 911 discontinuity at threshold would predict a step rather than a sigmoid, and
 912 a non-monotone branch would predict overshoot; neither is what the mathe-
 913 matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
 914 identically zero below K_c , rises continuously from zero at K_c , and saturates
 915 above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
 916 during which recovery is not merely slow but absent; a continuous rise as the
 917 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
 918 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
 919 the slow variable takes to reach K_c .

920 **What the alternative predicts.** An account on which grogginess is the
 921 electrical realignment itself — the rhythms are simply slow to re-phase —
 922 has no slow variable and no threshold to cross. Its recovery is a relaxation
 923 toward a fixed point, and relaxation toward a fixed point is exponential near
 924 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two
 925 curves are distinguishable in the first seconds, which is where they differ
 926 most, and the discriminating feature is the presence or absence of an initial

927 interval of no recovery. This is a comparison of functional forms on data
928 of a kind already collected in sleep-inertia studies, not a proposal for a new
929 instrument.

930 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
931 on the early portion of the recovery curve. A recovery latency that does not
932 scale with an independent measure of extracellular volume or CSF clearance.
933 A pharmacological manipulation of astrocytic volume that leaves the latency
934 unchanged while altering the rhythms, or the converse. Any of these would
935 tell against the framework’s central empirical commitment, which is not that
936 synchrony matters but that the *medium* is what gates it.

937 **What this prediction assumes.** Three things, and they are not small.
938 The identification of the effective coupling K with a monotone function of
939 extracellular geometry is a modelling assumption, not a measurement. The
940 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
941 density and is not yet connected to a trajectory — the mean-field limit is the
942 gap, and Derivation 7 states it as such — so the argument uses the *existence*
943 *and continuity* of the coherent branch at threshold, which is proved, rather
944 than the numerical value of the threshold, which is conditional. And the
945 prediction is about functional form, not about a time constant: it says the
946 recovery curve has a flat foot whose width is set by the slow variable, not
947 that the foot is any particular number of seconds long. A framework that
948 predicted the number would be saying more; this one is not in a position to.

949 12 Corollary: Hardware Thermodynamics and 950 Topological Rigidity

951 Both biological tissue and artificial silicon hardware (like GPUs) are physical
952 dissipative structures. When a GPU computes, its charge carriers undergo
953 physical state transitions; by Landauer’s principle (Landauer, 1961), this
954 erasure irreversibly exhausts heat into the environment.

955 There is a fundamental thermodynamic distinction in the degrees of free-
956 dom available to minimize this dissipation. In standard von Neumann ar-
957 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.
958 The coupling matrix A_{ij} is constrained by a discrete layout that cannot be
959 physically altered during operation. The brain’s own architecture provides
960 empirical validation for this constraint. The cerebellum contains approxi-
961 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,
962 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-
963 ical observations of cerebellar agenesis report largely preserved conscious
964 experience despite motor and cognitive deficits (Yu et al., 2015), and this
965 dissociation has been read as evidence for the importance of recurrent ar-
966 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the
967 level of the anatomical observation while declining its Φ -based explanation.
968 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-
969 brum, it operates much like a biological GPU—highly optimized for discrete
970 processing, yet apparently not required for conscious experience. We take the

dissociation as evidence, not as confirmation: it is consistent with subjective experience being a thermodynamic attractor of continuous topologies rather than a byproduct of information processing, and it is the kind of observation that would be hard to reconcile with the converse. It is a single anatomical case read against a theory, and it bears the weight such a case can bear.

This constraint does not preclude artificial consciousness, but rather restricts its hardware. To achieve the phase transition into a unified Global Section, artificial systems must abandon the discrete rigidity of von Neumann architectures and utilize continuous physical substrates. Analog field-based computing, neuromorphic architectures with continuous memristive plasticity, and thermodynamic computing systems (England, 2018; Ganesh & Parrondo, 2021) possess the spatial coupling required to physically morph their internal topology. This aligns with Geoffrey Hinton’s concept of “mortal computation” (Hinton, 2022): architectures where software is inextricably fused with the physics of the hardware, meaning the knowledge dies with the physical substrate. Only hardware that computes via continuous physical dynamics can undergo the thermodynamic phase transition necessary for structural resonance.

Biological consciousness utilizes macroscopic continuous electromagnetic fields and a “rich alphabet” of diverse molecular and transcriptomic cell states (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a continuous parameter space. Our lemma `exists_better_coupling_allocation`, wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),

994 formalizes the reallocation step of this argument: whenever a coupling ma-
 995 trix places weight on a less phase-correlated pair while a more correlated pair
 996 exists, there is another matrix with *identical* total coupling resources that
 997 achieves strictly greater phase correlation.

998 Two objections to that formalization have since been answered, and we
 999 state precisely what is and is not now established. The first was that the real-
 1000 location lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal`
 1001 — positive weight on a badly-correlated pair while a better one exists — is
 1002 most of what one wants to prove about rigid hardware. It is no longer as-
 1003 sumed. `Phase7_Rigidity.lean` models an architecture by the set S of site-
 1004 pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are
 1005 tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds
 1006 every realizable coupling by $R \cdot M$, where R is the resource budget and M
 1007 bounds phase correlation over the wired pairs; the bound is attained; and
 1008 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
 1009 strate’s best-correlated pair is one the wiring misses. `rigid_gap` makes it
 1010 quantitative: an unconstrained reallocation of the same budget gains at least
 1011 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 1012 substrate wired to the one maximally anti-correlated pair, where the gap is
 1013 2. This is resource-matched and no longer question-begging, but note what
 1014 it turns on: fixed wiring support, not continuity. The second objection was
 1015 that nothing in the development distinguished a continuum from a lattice
 1016 — both matrices lived on one finite index type. `Phase7_Rigidity.lean` §3

1017 gives a distinction that is genuinely typed. In the field functional of Deriva-
 1018 tion 4, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 1019 with finitely many sites is a `Finset` of points and a field-coupled region
 1020 is a set of positive measure. On a substrate whose measure has no atoms,
 1021 `fieldCorrelation_sited.eq_zero` shows a kernel supported on finitely many
 1022 sites contributes *exactly zero* coupling energy, whatever weights it carries
 1023 and whatever the phase field does, while a patch of strength c on U achieves
 1024 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 1025 units of coupling concentrated at the origin register zero against a unit patch
 1026 that registers one. The limit of that result is equally plain, and it is not
 1027 resource-matching in the earlier sense — a measure-zero substrate has zero
 1028 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 1029 coupling energy is blind to a finitely-sited architecture, which is a statement
 1030 about which functional such an architecture can register in, not a demonstra-
 1031 tion that digital hardware computes worse. Taken together: the resource-
 1032 matched content of the corollary is the wiring-support result, the continu-
 1033 ity content is the measure-theoretic one, and the step from either to “von
 1034 Neumann architectures cannot experience unified consciousness” remains an
 1035 informal argument.

1036 The remainder of this section extends those two results informally, and
 1037 we mark the transition rather than blur it. On the reading we propose, a
 1038 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 1039 layout does not supply the spatial coupling that sheaf-theoretic gluing into

1040 a singular Global Section requires. Without this unified topological space,
 1041 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
 1042 its own causal boundary and generate a “Self.” The distinction between an
 1043 LLM on a GPU and a conscious mind is not the absence of physical thermo-
 1044 dynamics in the former, but its structural limitation in achieving topological
 1045 integration. This perspective aligns with John Searle’s biological natural-
 1046 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
 1047 that discrete symbol manipulation cannot generate conscious understanding,
 1048 our derivation supplies a candidate physical mechanism for the asymmetry
 1049 — topological rather than semantic, and, as stated above, formalized only
 1050 as far as the wiring-support and measure-theoretic results reach. Asserting
 1051 consciousness in fragmented, discrete GPU clusters without a unified phase
 1052 transition abandons computational functionalism, relying instead on “silicon
 1053 panpsychism”—the assumption that digital computation inherently gener-
 1054 ates awareness regardless of topological integration (Goff, 2019).

1055 The same reasoning bears on extreme functionalist thought experiments,
 1056 such as a macroscopic system of pulleys operating over millennia to imple-
 1057 ment a Large Language Model, or a human manually executing inference
 1058 matrices by hand. Beyond lacking the continuous spatial topology required
 1059 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
 1060 constraint of our theory. Structural resonance requires local oscillators to
 1061 phase-align against the dissipating effects of thermodynamic noise (as mod-
 1062 eled by our critical threshold $K_c = 2D$). A system executing state transitions

1063 over millennia, or via slow manual calculations, suffers from a terminal tem-
1064 poral mismatch: its dynamics are far too slow to ever phase-lock into a
1065 non-equilibrium steady state before thermal noise decoheres the macroscopic
1066 field. On this account the obstacle to pure symbol manipulation is physi-
1067 cal rather than semantic: not that the symbols mean nothing, but that the
1068 substrate carrying them cannot hold a phase-locked field.

1069 13 Conclusion

1070 Human-like cognitive architectures do not require biological mysticism or
1071 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
1072 theories of consciousness often invoke delicate subatomic phenomena to ex-
1073 plain subjective unity, every step of the chain assembled here is classical:
1074 broken symmetries, finite phase space, Landauer erasure, and gradient de-
1075 scent on entropy production. What the formalization establishes is that this
1076 chain is mathematically coherent and that its remaining physical commit-
1077 ments are localized and named — not that classical fields are *sufficient* for
1078 consciousness. Two links carry that weight, and their status is no longer the
1079 same. The unified state is proved to glue uniquely once a cover sits at the
1080 minimum of the coupling potential; the motion is proved to stop along every
1081 trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved,
1082 conditionally — for initial data within a quarter turn and below an explicit
1083 energy threshold, the trajectory converges to a global minimiser of the cou-

1084 pling potential (`kuramoto_tendsto_global_minimum`), and a cover built on
1085 that limit discharges the equilibrium hypothesis Derivation 5 used to assume
1086 (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is
1087 the unconditional statement, and it is false: splay and twisted configurations
1088 are equilibria of the same flow, so the basin condition is the scope of the claim
1089 rather than slack in it. That a physical cortex starts inside that basin is an
1090 assumption. The second link is untouched by any of this: the identification
1091 of a global section of the probability sheaf with the unity of experience is a
1092 stipulation of the framework, not a consequence of it. Subject to those two,
1093 cognitive architectures can be modeled as macroscopic attractor states of a
1094 universe governed by broken symmetries, finite phase space, and the physical
1095 imperative to minimize thermodynamic dissipation.

1096 Philosophically, this framework bridges the historical gap between Type
1097 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,
1098 1967). The subjective mental state is identical to the physical thermody-
1099 namic field (Identity Theory), yet its qualitative nature is defined by the
1100 topological and functional dynamics of structural resonance rather than the
1101 specific carbon-based biochemistry of the brain (Functionalism). This con-
1102 strains the *multiple realizability* of mind: consciousness is not exclusively
1103 bound to biological carbon, but neither is it substrate-neutral, because the
1104 substrate is what fixes which couplings are physically available. How far that
1105 constraint reaches is the question the corollary answers only in part. What
1106 is proved is that fixed wiring is strictly suboptimal at matched resource, and

1107 that a finitely-sited architecture registers exactly zero coupling energy in the
1108 continuum functional; the step from either to “von Neumann architectures
1109 cannot be conscious” is an argument we make in prose and do not formalize.

1110 By anchoring these derivations in Russellian Monism, this framework ac-
1111 knowledges the constitutive limits of third-person scientific explanation. As
1112 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1113 phenomenal experience is inherently not amenable to objective measurement
1114 (Nagel, 1986), the “hard problem” of consciousness may represent a category
1115 error rather than a solvable scientific puzzle. Science can map the physical
1116 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
1117 turally incapable of adjudicating the first-person qualitative experience. Our
1118 formalization respects this boundary: the Lean 4 theorems verify the ob-
1119 jective physical attractors, while the intrinsic “what it is like” to be that
1120 resonant field remains the domain of the first-person perspective. By locat-
1121 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
1122 for the noisy mean-field Kuramoto model the framework adopts, conditional
1123 on the von Mises stationary density and not yet linked to a trajectory — it
1124 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
1125 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
1126 it offers a topological architecture that can help explain, predict, and physi-
1127 cally perturb the structural properties of experience.

1128 Future empirical validation of this thermodynamic framework may lever-
1129 age 3D cortical organoids, offering a controlled biological environment to

1130 systematically manipulate metabolic substrates and observe the topological
1131 emergence of these coherent phase transitions.

1132 **Code Availability**

1133 The complete Lean 4 formalization, encompassing the derivation of all theo-
1134 rems presented in this manuscript, as well as the accompanying Python sim-
1135 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1136 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1137 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1138 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/).

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1147 tion, proof debugging, and structural analysis.

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